

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12396404
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Childs; Albert

Repositionable windguard assembly for agricultural baler feeder system

Abstract

An agricultural machine includes: a machine frame; and a feeder system coupled with the machine frame and including a windguard assembly, the windguard assembly being configured for selectively occupying a first position and a second position, the first position being associated with a normal flow direction of a crop material relative to the feeder system, the second position being associated with a reversal of the normal flow direction of the crop material.

Inventors:	Childs; Albert (Meadville, PA)
Applicant:	CNH Industrial America LLC (New Holland, PA)
Family ID:	1000008782905
Assignee:	CNH Industrial America LLC (New Holland, PA)
Appl. No.:	17/589387
Filed:	January 31, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230276738 A1	Sep. 07, 2023

Publication Classification

Int. Cl.: A01F15/10 (20060101); A01F15/07 (20060101)

U.S. Cl.:

CPC A01F15/106 (20130101); A01F15/07 (20130101); A01F2015/107 (20130101)

Field of Classification Search

CPC: A01F (15/106); A01F (15/07); A01F (2015/107); A01F (15/10); A01D (89/008)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2481244	12/1948	Russell	198/817	A01F 15/10
2872772	12/1958	Nolt	56/364	A01D 89/008
3924391	12/1974	Cheatum	56/364	A01F 15/08
4411127	12/1982	Diederich, Jr. et al.	N/A	N/A
4516389	12/1984	Core	56/190	A01F 15/106
6295797	12/2000	Naaktgeboren et al.	N/A	N/A
6877304	12/2004	Smith	56/364	A01F 15/106
7107748	12/2005	McClure	56/190	A01D 89/008
7650741	12/2009	Graber et al.	N/A	N/A
7654069	12/2009	Dunham et al.	N/A	N/A
8751115	12/2013	Derscheid	N/A	N/A
9681603	12/2016	McClure et al.	N/A	N/A
10111387	12/2017	Derscheid	N/A	A01D 89/008
11547053	12/2022	Lammerant	N/A	A01D 89/008
11700793	12/2022	Tacke	100/142	A01F 15/101
12063892	12/2023	Babler	N/A	A01D 75/187
12089536	12/2023	Childs	N/A	A01D 89/008
2006/0277888	12/2005	Erdmann	56/344	A01D 89/008
2008/0163600	12/2007	Schrag	56/341	A01D 89/008
2008/0264028	12/2007	Woodford	N/A	N/A
2013/0167500	12/2012	Kuhn	56/364	A01D 89/002
2014/0250855	12/2013	Vandamme	56/379	A01D 89/008
2021/0274716	12/2020	Lammerant et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102008041331	12/2009	DE	A01D 89/008
0339733	12/1988	EP	N/A
0403899	12/1989	EP	A01D 75/187
2067399	12/2008	EP	N/A
2156729	12/2009	EP	A01D 89/008
2777379	12/2015	EP	A01D 34/001
3912456	12/2020	EP	N/A

OTHER PUBLICATIONS

Extended European Search Report for EP Application No. 23153957.8 dated Jul. 5, 2023 (seven pages). cited by applicant

Primary Examiner: Fabian-Kovacs; Arpad

Attorney, Agent or Firm: CM Law PLLC

Background/Summary

FIELD OF THE INVENTION

(1) The present invention pertains to an agricultural baler, and, more specifically, to a windguard assembly of the agricultural baler.

BACKGROUND OF THE INVENTION

(2) Agricultural harvesting machines, such as agricultural balers (which can be referred to as balers), have been used to consolidate and package crop material (which can also be referred to as forage, forage material, or forage crop material) so as to facilitate the storage and handling of the crop material for later use. Often, a mower-conditioner cuts and conditions the crop material and forms it into a swath or windrow to dry in the sun. When the cut crop material is properly dried (depending upon the application), an agricultural harvesting machine, such as an agricultural baler, travels along the swath or windrows (hereinafter, collectively referred to as windrows, unless otherwise specified) to pick up the crop material. Agricultural balers can be, for example, round balers, large square balers, and small square balers, which form the bales into corresponding shapes, as is known.

(3) More specifically, with respect to round balers, pickups of the baler gather the cut and windrowed crop material from the ground, and then convey the cut crop material toward a bale-forming chamber within the baler (that is, the bale chamber). A drive mechanism operates to activate any pickups of a pickup assembly, augers, and/or a rotor (which can be referred to as a feeder) of a rotary feeder assembly of a feeder system. A pickup can convey crop material in an overshot manner, while a rotor can convey crop material toward or into the bale chamber in an undershot manner. A conventional bale chamber of a round baler may include a pair of opposing

sidewalls with a series of rolls (which can be referred to as rollers) and belts that rotate and compress the crop material into a cylindrical shape. When the bale has reached a desired size and density, a wrapping assembly, which includes wrap material, may wrap the bale to ensure, at least in part, that the bale maintains its shape and density. The wrap material can include a film (such as a flexible plastic wrap) or a net (which can be referred to as net wrap). For example, wrap material may be used to wrap the bale of crop material. After wrapping, a cutting or severing mechanism of the wrapping assembly may be used to cut the wrap material once the bale has been wrapped. The wrapped bale may be ejected from the baler and onto the ground by, for example, raising a tailgate of the baler. The tailgate is then closed, and the cycle repeated as necessary and desired to manage the field of cut crop material.

(4) The feeder system can include not only the pickup assembly and the rotor assembly, for example, but also a floor, a cutting assembly, and a windguard assembly. The pickup includes a plurality of tines to pick up the crop material disposed in a windrow. The rotor, which is downstream of the pickup, can be positioned above the floor which the crop material traverses prior to entering the bale chamber and can work in conjunction with, and cooperatively with, the cutting assembly. The rotor can include a rotor shaft (extending transversely) and a plurality of tines (which can have a generally triangular or star-shaped configuration) spaced apart across the transverse extent of the baler. The tines are configured to engage and thereby to push the crop material towards the bale chamber and can be grouped in pairs, with a relatively short distance between the tines of a given pair. The floor can include a plurality of slots across the transverse extent of the floor, each slot extending longitudinally in the floor (that is, parallel or otherwise aligned with a direction of crop flow). The cutting assembly can include a plurality of knives (which can also be referred to as cutters) which are selectively received in the slots of this floor, respectively. When inserted through the slots so that the knives extend at least partially above the floor, each respective knife (depending upon the design) can extend between a pair of tines of the rotor, as the rotor shaft rotates the tines. Further, the knives, as they extend through the slots above the floor are configured to cut the crop material to a predetermined length, as the crop material passes by the knives prior to the crop material entering the bale chamber. These knives can be configured to drop at least partially below the floor upon being struck by a foreign object. The windguard assembly can include a roller and a plurality of tines extending aft and can be configured to retain the crop material along at least a part of its flow path as it proceeds rearward by the pickup and the rotor to the bale chamber and, in so doing, can be positioned adjacent to an upper portion of the crop material. Further, the windguard assembly can be pivotably attached to the pickup assembly.

(5) Occasionally, such as with round balers, the crop material becomes plugged in the feeder system. That is, the crop material bunches up so as to form a plug of crop material (which can be referred to as a crop plug), which does not allow, or otherwise at least partially inhibits, the crop material from flowing from the feeder system into the bale chamber. Round balers prone to plugging can have a variable position (drop) floor below the rotor (also known as the feeder) that allows the crop plug to be passed, by increasing a space between the floor and the rotor. Such a drop floor can be effective in unplugging if only the rotor is plugged but often is not effective for unplugging when the crop material is plugged between the windguard assembly and the pickup (resulting in a pickup plug). Such a plug can result, for example, when an operator's ground speed has exceeded the feeding capacity of the baler. When a plug of crop material occurs that cannot be cleared by the drop floor, some operators may use a manually or remotely operated rotor reverser of a reversing system (so as to reverse a direction of the flow of the crop material), which can be useful in loosening the crop material but typically is unable to clear a pickup entirely. The reversing system cannot effectively clear a pickup plug because a conventional construction of windguard assembly geometry acts as a barrier, preventing the flow of the crop material in the reverse direction. For, the tines of the windguard assembly act as barbs, jamming the crop material and thus

stopping the reverse flow of crop material and preventing the crop from clearing. Manual removal of the plug, thus, is often necessary for a pickup plug, which often requires the operator to remove the windguard assembly and to pull the crop material out of the pickup assembly by hand—a timely, labor-intensive job which often occurs in uncomfortable field conditions (heat, bugs, dust).

(6) What is needed in the art is a way to reposition the windguard to facilitate removal of a plug of crop material from a feeder system when a drop floor is insufficient to do so, without having to clear the plug manually.

SUMMARY OF THE INVENTION

(7) The present invention provides an agricultural baler with a feeder system including a windguard assembly configured for selectively moving between a plurality of positions, such that a plug of crop material can be removed from the feeder system when the flow direction of the crop material in the feeder system is reversed.

(8) The invention in one form is directed to a windguard assembly of a feeder system of an agricultural baler, the agricultural baler including a baler frame, the feeder system being coupled with the baler frame, the windguard assembly including: the windguard assembly, which is configured for selectively occupying a first position and a second position, the first position being associated with a normal flow direction of a crop material relative to the feeder system, the second position being associated with a reversal of the normal flow direction of the crop material.

(9) The invention in another form is directed to an agricultural machine, including: a machine frame; and a feeder system coupled with the machine frame and including a windguard assembly, the windguard assembly being configured for selectively occupying a first position and a second position, the first position being associated with a normal flow direction of a crop material relative to the feeder system, the second position being associated with a reversal of the normal flow direction of the crop material.

(10) The invention in yet another form is directed to a method of using an agricultural baler, the method including the steps of: providing a baler frame and a feeder system, the feeder system being coupled with the baler frame and including a windguard assembly; occupying selectively, by the windguard assembly, a first position, the first position being associated with a normal flow direction of a crop material relative to the feeder system; and occupying selectively, by the windguard assembly, a second position, the second position being associated with a reversal of the normal flow direction of the crop material.

(11) An advantage of the present invention is that it provides a windguard design of variable geometry for use with a harvester pickup to allow effective rejection of a crop plug from the machine.

(12) Another advantage is that it provides that, in varying the geometry of the windguard, the windguard, or at least a portion thereof, can be extended forward to open up space for the crop plug to be removed.

(13) Yet another advantage is that it provides that, in varying the geometry of the windguard, the tines of the windguard can be pivoted so that they do not act as barbs when the plug is attempted to be removed.

(14) Yet another advantage is that the windguard assembly does not need to be removed in order to remove a crop plug.

(15) Yet another advantage is that a variation of the geometry of the windguard assembly can be controlled by a controller, based upon sensor feedback, in order to optimize crop feeding.

Alternatively, the variation of the geometry of the windguard assembly can be controlled by a user.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For the purpose of illustration, there are shown in the drawings certain embodiments of the present invention. It should be understood, however, that the invention is not limited to the precise arrangements, dimensions, and instruments shown. Like numerals indicate like elements throughout the drawings. In the drawings:

(2) FIG. 1 illustrates a schematic side view of an exemplary embodiment of an agricultural vehicle, formed as a tractor, and an agricultural baler, the agricultural baler including a feeder system including a windguard assembly, in accordance with an exemplary embodiment of the present invention;

(3) FIG. 2 illustrates a perspective view of the feeder system of the agricultural baler of FIG. 1 (at least partially schematically), the feeder system including the windguard assembly in a first position, with portions broken away, in accordance with an exemplary embodiment of the present invention;

(4) FIG. 3 illustrates a side view of the feeder system of FIG. 2 (at least partially schematically), the windguard assembly being in the first position, with portions broken away;

(5) FIG. 4 illustrates a side view of the feeder system of FIG. 2 (at least partially schematically), the windguard assembly being in the second position, with portions broken away;

(6) FIG. 5 illustrates a perspective view of another exemplary embodiment of the feeder system of the agricultural baler (at least partially schematically), the feeder system including the windguard assembly in the first position, with portions broken away, in accordance with an exemplary embodiment of the present invention;

(7) FIG. 6 illustrates a side view of the feeder system of FIG. 5 (at least partially schematically), the windguard assembly being in the second position, with portions broken away; and

(8) FIG. 7 illustrates a flow diagram showing a method of using an agricultural baler, in accordance with an exemplary embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(9) The terms “forward”, “rearward”, “left” and “right”, when used in connection with the agricultural vehicle, agricultural baler, and/or components thereof are usually determined with reference to the direction of forward operative travel of the agricultural vehicle and/or agricultural baler, but they should not be construed as limiting. The terms “longitudinal” and “transverse” are determined with reference to the fore-and-aft direction of the agricultural vehicle and/or agricultural baler and are equally not to be construed as limiting. The terms “downstream” and “upstream” are determined with reference to the intended direction of crop material flow during operation, with “downstream” being analogous to “rearward” and “upstream” being analogous to “forward.”

(10) Referring now to the drawings, and more particularly to FIG. 1, there is shown an embodiment of an agricultural vehicle **100** (which can be referred to as a work vehicle **100**) towing an agricultural baler **101** (any baler herein can be referred to as an agricultural machine), in accordance with the present invention, to perform a baling operation within a field (FIG. 1 is generic to all embodiments of the present invention discussed herein). As shown, work vehicle **100** can be configured as an agricultural tractor, such as an operator-driven tractor or an autonomous tractor. However, in some embodiments, the work vehicle **100** may correspond to any other suitable vehicle configured to tow a baler across a field or that is otherwise configured to facilitate the performance of a baling operation, including an autonomous baling vehicle. Additionally, as shown, baler **101** can be configured as a round baler configured to generate round bales. However, in some embodiments, baler **101** may have any other suitable configuration, including being configured to generate square or rectangular bales. It should be further appreciated that baler **101**, while shown as being towed by tractor **100**, may also be a self-propelled baler that does not rely on a separate vehicle for propulsion and/or power to function. While the present invention is described

with respect to a baler, it can be appreciated that other the present invention can be used in conjunction with other agricultural machines, such as vehicles and/or implements, such as a self-loading forage wagon.

(11) Work vehicle **100** includes a pair of front wheels **102**, a pair of rear wheels **103**, and a chassis **104** coupled to and supported by the wheels **102**, **103**. An operator's cab **105** may be supported by a portion of the chassis **104** and may house various input devices for permitting an operator to control the operation of work vehicle **100** and/or baler **101**. Additionally, work vehicle **100** may include an engine and a transmission mounted on chassis **104**. The transmission may be operably coupled to the engine and may provide variably adjusted gear ratios for transferring engine power to wheels **103** via a drive axle assembly.

(12) As shown in FIG. 1, work vehicle **100** may be coupled to baler **101** via a power take-off (PTO) **106** and a tongue **107** to a hitch of work vehicle **100** to allow vehicle **100** to tow baler **101** across the field. As such, work vehicle **100** may, for example, guide baler **101** toward crop material **136** deposited in windrows on the field. As is generally understood, to collect the crop material **136**, baler **101** includes a feeder system **108** (which can be referred to generally as a crop collector) mounted on a front end of baler **101**. Feeder system **108** may, for example, include a pickup assembly **111**, a rotary feeder assembly **121**, a floor **131**, and a windguard assembly **128**. Pickup assembly **111** includes a rotating wheel with tines that collect crop material **136** from the ground and direct crop material **136** toward a bale chamber **109** of baler **101** in an overshot manner (rotating clockwise in FIG. 1). Rotary feeder assembly **121** includes a rotating shaft (rotor shaft) and a plurality of rotor devices **144** (which can be referred to as rotors **144**) mounted to the shaft so as to rotate about the shaft in a counter-clockwise manner (as seen in FIG. 1 and shown by arrow **134**), rotors **144** configured to push or otherwise to move crop material **136** towards or into bale chamber **109**, in an undershot manner. Feeder system **108** can also include a rotating shaft (not shown in FIG. 1) generally between pickup assembly **111** and rotary feeder assembly **121** that includes side augers for moving crop material **136** inwardly prior to entering bale chamber **109**. FIG. 1 shows crop material **136** lying in a windrow on the ground of the field and being picked up by pickup assembly **111** of feeder system **108**. Upon being picked up, crop material flow **136** flows over pickup assembly **111**, under windguard assembly **128**, and under rotary feeder assembly **121**, as indicated by arrow **132**. Arrow **132** thus indicates the normal flow direction **132** of crop material **136** relative to feeder system **108**. Conversely, arrow **133** shows a flow direction which is opposite the normal flow direction of crop material **136**; this opposite flow direction **133** can also be referred to as a reverse flow direction **133** (that is, a reversal of normal flow direction **132**). FIG. 1 shows baler **101** schematically and thus with one pair of wheels, though it can be appreciated that baler **101** can include one or more additional pair of wheels, as in FIG. 2.

(13) Inside bale chamber **109**, rollers, belts, and/or other devices compact the crop material **136** to form a generally cylindrically-shaped bale **110**. Bale **110** is contained within baler **101** until ejection of bale **110** is instructed (e.g., by the operator and/or a baler controller **123** of baler **101**). In some embodiments, bale **110** may be automatically ejected from baler **101** once bale **110** is formed, by baler controller **123** detecting that bale **110** is fully formed and outputting an appropriate ejection signal. Further, work vehicle **100** includes a control system **114**, which includes a controller **115**, which includes a processor **116**, memory **117**, data **118**, and instructions **119**. Control system **114** can further include an input/output device **120** such as a laptop computer (with keyboard and display) or a touchpad (including keypad functionality and a display), device **120** being configured for a user to interface therewith.

(14) As shown in FIG. 1, baler **101** may also include a tailgate **112** movable between a closed position (as shown in the illustrated embodiment) and an opened position via a suitable actuator assembly. Tailgate **112** and/or the actuator assembly may be controlled to open and close by baler controller **123**. In the closed position, tailgate **112** may confine or retain bale **110** within baler **101**. In the open position, tailgate **112** may rotate out of the way to allow bale **110** to be ejected from the

bale chamber **109**. Additionally, as shown in FIG. 1, baler **101** may include a ramp **113** extending from its aft end that is configured to receive and direct bale **110** away from baler **113** as it is being ejected from bale chamber **109**. In some embodiments, ramp **113** may be spring loaded, such that ramp **113** is urged into a raised position, as illustrated. In such embodiments, the weight of bale **110** on ramp **113** may drive ramp **113** to a lowered position in which ramp **113** directs bale **110** to the soil surface. Once bale **110** is ejected, bale **110** may roll down ramp **113** and be deposited onto the field. As such, ramp **113** may enable bale **110** to maintain its shape and desired density by gently guiding bale **110** onto the field. Further, baler **101** includes a control system **122**, which includes controller **123**, which includes a processor **124**, memory **125**, data **126**, and instructions **127**. Controller **123** can communicate with controller **115**, so that controller **115** outputs information to the display of input/output device **120** of work vehicle **100**, thereby informing a user of various conditions of baler **101** and bales **110** forming or formed therein. Further, baler **101** includes a frame **130** (which can be referred to as baler frame **130**, or more generally as a machine frame **130**) to which all of the components of baler **101** are directly or indirectly coupled. Thus, feeder system **108** and thus also pickup assembly **111**, rotary feeder assembly **121**, and windguard assembly **128** are coupled with frame **130**.

(15) It should be appreciated that the configuration of work vehicle **100** described above and shown in FIG. 1 is provided only as one example. Thus, it should be appreciated that the present disclosure may be readily adaptable to any manner of work vehicle configuration. For example, in an alternative embodiment, a separate frame or chassis may be provided to which the engine, transmission, and drive axle assembly are coupled, a configuration common in smaller tractors. Still other configurations may use an articulated chassis to steer work vehicle, or rely on tracks in lieu of wheels **102**, **103**. Additionally, as indicated previously, work vehicle **100** may, in some embodiments, be configured as an autonomous vehicle. In such embodiments, work vehicle **100** may include suitable components for providing autonomous vehicle operation and, depending on the vehicle configuration, need not include the operator's cab **105**.

(16) Additionally, it should be appreciated that the configuration of baler **101** described above and shown in FIG. 1 is provided only as one example. Thus, it should be appreciated that the present disclosure may be readily adaptable to any manner of baler configuration, or other agricultural machines, such as a vehicle and/or implement, such as a self-loading forage wagon. For example, as indicated previously, baler **101** may, in some embodiments, correspond to a square baler configured to generate square or rectangular bales. It should be further appreciated that the illustration of baler **101** in FIG. 1 is schematic.

(17) Further, in general, controllers **115**, **123** may each correspond to any suitable processor-based device(s), such as a computing device or any combination of computing devices. Each controller **115**, **123** may generally include one or more processor(s) **116**, **124** and associated memory **117**, **125** configured to perform a variety of computer-implemented functions (e.g., performing the methods, steps, algorithms, calculations and the like disclosed herein). Thus, each controller **115**, **123** may include a respective processor **116**, **124** therein, as well as associated memory **117**, **125**, data **118**, **126**, and instructions **119**, **127**, each forming at least part of the respective controller **115**, **123**. As used herein, the term “processor” refers not only to integrated circuits referred to in the art as being included in a computer, but also refers to a controller, a microcontroller, a microcomputer, a programmable logic controller (PLC), an application specific integrated circuit, and other programmable circuits. Additionally, the respective memory **117**, **125** may generally include memory element(s) including, but not limited to, computer readable medium (e.g., random access memory (RAM)), computer readable non-volatile medium (e.g., a flash memory), a floppy disk, a compact disc-read only memory (CD-ROM), a magneto-optical disk (MOD), a digital versatile disc (DVD), and/or other suitable memory elements. Such memory **117**, **125** may generally be configured to store information accessible to the processor(s) **116**, **124**, including data **118**, **126** that can be retrieved, manipulated, created, and/or stored by the processor(s) **116**, **124** and the

instructions **119**, **127** that can be executed by the processor(s) **116**, **124**. In some embodiments, data **118**, **126** may be stored in one or more databases.

(18) Baler controller **123**, herein, is assumed to be the primary controller for controlling operations of baler **101**. However, controller **123** of baler **101**, as indicated in FIG. 1, can be in communication with controller **115** of work vehicle **100**, thereby forming a unified control system **129**, such that any or all information associated with either controller **115**, **123** can be shared with the other controller **115**, **123**, and either controller **115**, **123** can perform the functions of the other controller **115**, **123**. Controllers **115**, **123** can communicate with each other in any suitable manner, such as a wired connection or a wireless connection, such as radio signals (RF), light signals, cellular, WiFi, Bluetooth, Internet, via cloud-based devices such as servers, and/or the like. Further, while not shown, both controllers **115**, **123** can communicate with a remotely located data center (which can also be a part of unified control system **129**), which controllers **115**, **123** can communicate with by any suitable way, such as those just referenced. Such a data center can include its own controller (and thus processor(s), memory, data, and instructions, substantially similar to that described above with respect to controllers **115**, **123**) which can be configured to perform any of the functions associated with controllers **115**, **123**. Controllers **115**, **123** and the data center can be a part of any network facilitating such communication therebetween, such as a local area network, a metropolitan area network, a wide area network, a neural network, whether wired or wireless.

(19) Referring now to FIG. 2, there is shown feeder system **108** of baler **101**, feeder system **108** (shown in part schematically) including pickup assembly **111**, rotary feeder assembly **121**, and windguard assembly **128**. Pickup assembly **111** includes a plurality of pickup tines **240** which are configured to pick up the windrow of crop material **136** from the ground as tines **240** rotate in a counter-clockwise direction (as viewed from the bottom left of the page of FIG. 2), with crop **136** moving in a normal flow direction **132** once picked up by tines **240**. Pickup assembly **111** further includes a frame **241** (which can be referred to as a pickup frame **241**) including a windguard mounting device **242** on each lateral side of feeding system **108**, which can be fastened to other portions of pickup frame **241**, such as by way of rivets or threaded fasteners, such as bolts or screws, or the like. Windguard mounting device **242** further includes a stop mechanism **243** configured to selectively receive thereon a portion of windguard assembly **128**; windguard mounting device **242** can be attached to pickup frame **241** in any suitable manner, such as by way of welding or by way of fasteners. Windguard frame **241** and windguard mounting device **242** can each be made of any suitable material, such as steel or a polymer. Rotary feeder assembly **121** includes rotors **144**, each of which includes a plurality of generally triangular shaped tines, the rotors **144** being configured to move crop material **136** towards bale chamber **109** as rotors **144** rotate in a clockwise direction (as viewed from the bottom left of the page of FIG. 2).

(20) Windguard assembly **128** is coupled with pickup assembly **111** and, according to an exemplary embodiment of the present invention, includes a first arm **245** (on each lateral side of feeding system **108**), a second arm **246** (on each lateral side of feeding system **108**), a roller **247**, a tine mechanism **248**, a chain assembly **249**, a linkage assembly **250** (on each lateral side of feeding system **108**), and an actuator **251** (on each lateral side of feeding system **108** as shown in FIG. 2, though optionally only one such actuator **251** may be included). Windguard assembly **128** can be substantially similar on both lateral sides of a midline running longitudinally, such that a description of one lateral side of windguard assembly **128** substantially serves as a description of the opposing lateral side of windguard assembly **128**. Windguard assembly **128** is configured for selectively occupying a first position **237** and a second position **438**, the first position **237** being associated with normal flow direction **132** of a crop material **136** relative to the feeder system **108**, the second position **438** being associated with a reversal of the normal flow direction **132** of the crop material **136** (and thus a reversed flow direction **133**). In general, the first position **237** is when windguard assembly **128** occupies, at least in part, a rearward position (as in FIG. 2), and the second position **438** is when windguard assembly **128** occupies, at least in part, a forward position

(as in FIG. 4).

(21) First arm **245** (which can be referred to as a fore member or forearm) is pivotably coupled at connection **265A** with windguard mounting device **242** of pickup assembly **111** at a proximal end **352** of first arm **245** (stated another way, first arm **245** is supported by a pivot mount on windguard mounting device **242** of pickup assembly **111**). In FIG. 2, first arm **245** is shown in an intermediate position when second arm **246** is retracted and resting on stop mechanism **243**. Connection **265A** can be made by any suitable connectors and/or fasteners, such as rivets, pins, screws, nuts, washers, bearings, and/or bolts. Further, first arm **245** can define a first arm channel **253** as shown in FIG. 2, which can be substantially U-shaped. Side walls of first arm channel **253** can each define a slot **254**, which can be arcuate, for slidably receiving therein sliding structure, such as a pin or shaft of a connector or fastener, which can be part of connection **265H** (below). First arm **245** can be made of any suitable material, such as steel, a polymer, or carbon fiber, and can be made in any suitable manner, such as stamping, metal forming, welding, and/or injection molding.

(22) Second arm **246** (which can be referred to as a rear member or rear member) is pivotably coupled with first arm **245** at connection **265B** (at a distal end **355** of first arm **245** and a proximal end of second arm **246**) and can define, at least in part, a second arm channel **256**. Second arm **246** is also pivotably coupled with opposing ends of windguard roller **247** at connection **265C** (at a distal end of second arm **246**). Connections **265B** and **265C** can be made by any suitable connectors and/or fasteners, such as rivets, pins, screws, nuts, washers, bearings, and/or bolts. Second arm **246** can be made of any suitable material, such as steel, a polymer, or carbon fiber, and can be made in any suitable manner, such as stamping, metal forming, welding, and/or injection molding. Roller **247** extends between opposing second arms **246** across a front portion of windguard assembly **128** and can pivot in either direction, as is known, as crop material **136** contacts roller **247**. Though the windguard assembly of the present invention is discussed as having a roller, it can be appreciated that the present invention is not limited to a type of windguard assembly having a roller (and thus includes windguard assemblies without a roller).

(23) Tine mechanism **248** includes a bar **259** (which can also be referred to as a tube **259**, or tine bar **259**) and a plurality of tines **260** rigidly connected to bar **259**, tines **260** projecting generally in an aft direction. Bar **259** is pivotably coupled with second arm **246** at connection **265D**, such that bar can extend through each second arm **246** (this is different than a known prior art device, wherein a tine bar is welded to an arm). Connection **265D** can be made by any suitable connectors and/or fasteners, such as rivets, pins, screws, nuts, washers, bearings, and/or bolts. Tines **260** are configured for riding on a top portion of crop material **136** as crop material flows through feeder system **108** and thereby serves, at least in part, to retain crop material **136** within feeder system **108** and on a proper path to bale chamber **109**. Tine mechanism **248** can be made of any suitable material, such as steel, a polymer, or carbon fiber, and can be made in any suitable manner, such as stamping, metal forming, welding, and/or injection molding.

(24) Chain assembly **249** can include one or more chains **261** (such as two, as shown in FIG. 2). Each chain **261** can extend between tine mechanism **248** and a front portion of baler **101** on baler frame **130** so as to be pivotal therewith (by way of a hook(s), for example) and thereby hold and thus support portions of windguard assembly **128** when not otherwise supported, for example, by actuator **251** and/or crop material **136**. More specifically, first and second arms **245**, **246** can receive support by way stop mechanism **243** (i.e., FIGS. 2, 4, 6), crop material **136** (i.e., flowing normally through feeder system **108**, as a plug, or as a plug being backed out of feeder system **108**), or chain **261**. With respect to chain **261**, chain **261** can suspend at least portions of windguard assembly **128** when windguard assembly **128** is not otherwise supported by stop mechanism **243** or crop material **136**; when chain **261** supports portions of windguard assembly **128** chain **261** can be taut, and when it does not chain **261** can be slack. Chain assembly **249** can also include a chain sleeve **262** surrounding at least a portion of each chain **261**, chain sleeve **262**, at least in part, keeping chain **261** substantially straight in the vicinity of the connection between chain **261** and

tine mechanism **248** and can help chain **261** from interfering with other nearby structure. Chain sleeve **262** can be a relatively rigid material, such as a plastic pipe, and or made of flexible material.

(25) Linkage assembly **250** includes a first linkage **263** and a second linkage **264** pivotably coupled with first linkage **263**. First linkage **263** (which can be referred to as a push rod **263**) can be pivotably coupled with first arm **245** at connection **265E** (at a proximal end of first linkage **263**), which can be made by any suitable connectors and/or fasteners, such as rivets, pins, screws, nuts, washers, bearings, and/or bolts. First and second linkages **263**, **264** can be pivotably coupled together at connection **265F** (at a distal end of first linkage **263**), which can be made by any suitable connectors and/or fasteners, such as rivets, pins, screws, nuts, washers, bearings, and/or bolts. Second linkage **264** (which can be referred to as a crank arm **264**) can also be rigidly connected to bar **259**, in any suitable way, such as by welding. First and second linkages **263**, **264** can be made of any suitable material, such as steel, a polymer, or carbon fiber, and can be made in any suitable manner, such as stamping, metal forming, welding and/or injection molding. Further, linkage assembly **250** is configured for moving tine mechanism **248** between a down position **267** and an up position **468** when second arm **246** moves between a retracted position **257** and an extended position **458**, respectively, FIG. 2 showing tine mechanism **248** in the down position **267** and second arm **246** in the retracted position **257**. Thus, the present invention provides linkage assembly **250** which is configured for rotating and thereby for changing an angle of tines **260** dependent on pivot connection **265B**.

(26) Actuator **251**, as shown in FIG. 2, is pivotably coupled with first arm **245** and second arm **246** at connections **265G** and **265H**, respectively, each of which can be made by any suitable connectors and/or fasteners, such as rivets, pins, screws, nuts, washers, bearings, and/or bolts. Actuator **251** can be of any suitable configuration or type. For instance, actuator **251** can be a linear actuator or a rotary actuator. As a linear actuator, actuator can be a hydraulic cylinder assembly, a pneumatic cylinder assembly, or an electrically actuated cylinder assembly (which can include gearing and/or screws, for example). Herein, by way of example and not limitation, actuator **251** is shown and described as a linear actuator **251**, more specifically, a hydraulic cylinder assembly **251**, including a cylinder (which can also be referred to as a barrel), a piston within the cylinder, and a rod coupled with the piston. As one example, the cylinder can receive (in a portion of the cylinder) a hydraulic fluid (such as hydraulic oil), so as to cause the piston to be displaced within the cylinder and thereby to move the rod, causing the rod to extend; conversely, the hydraulic fluid can be caused to exit from a portion of the cylinder, so as to cause the piston to be displaced in an opposite direction within the cylinder and thereby to move the rod, causing the rod to retract. FIG. 2 shows the cylinder of actuator **251** including ports through which hydraulic fluid can enter and exit the cylinder, though additional structure of a hydraulic system (of which actuator **251** is a part) is not shown. Such a hydraulic system can include valving (for metering the proper amount of hydraulic fluid entering and exiting the cylinder), hydraulic lines, a reservoir of hydraulic fluid, and a pump, though other structures can be included or substituted therefor; structure of the hydraulic system can be maintained on baler **101** and/or work vehicle **100**. Connection **265G** is made with the cylinder of cylinder assembly **261**, and connection **265H** is made with the rod of cylinder assembly **251**. Actuator **251**, as shown in FIG. 2, can be positioned, at least in part, within first arm channel **253** and second arm channel **256**. As indicated above, first arm channel **253** includes side walls each defining an arcuate slot **254**, and the pin or shaft of the connector or fastener of connection **265H** connects the rod of cylinder assembly **251** not only with second arm **246** (pivotably) but also with first arm **245** by way of the opposing slots **254**, such that the rod of cylinder assembly **251** is slidably coupled with first arm **245** by way of slots **254**. Actuator **251** is configured for moving second arm **246** between the retracted position **257** of second arm **246** and the extended position **458** of second arm **246**, the retracted position **257** being associated with the first position **237** of windguard assembly **128**, the extended position **458** being associated with the second position **438**

of windguard assembly **128**. FIG. 2 shows windguard assembly **128** in the first position **237**, the rod of actuator **251** being extended to its farthest extent (such that the pin or shaft of connection **265H** is seated at a distal end of slot **254**), second arm **246** being in the retracted position **257**, and tines **260** being in their down position **267**. Thus, when actuator **251** is activated (i.e., by user, and/or by controller(s) **115**, **123**), second arm **246** is rotated relative to first arm **245** about an axis of rotation at connection **265B**. More specifically, when windguard assembly is positioned as shown in FIG. 2, retraction of the rod of actuator **251** causes second arm **246** to pivot counter-clockwise (viewed from the bottom left of the page of FIG. 2) about an axis of rotation at connection **265B**, causing extension forward of second arm **246**. This rotation of second arm **246** causes rotation of first linkage **263** at connection **265E**, first linkage **263** then pushing second linkage **264** such that first and second linkages **263**, **264** rotate at about an axis at pivot connection **265F** and second linkage rotates clockwise about an axis at connection **265D**, which causes tine bar **259** and tines **260** to rotate together clockwise to the up position **468**, thereby creating space for a plug of crop material **136** to be removed.

(27) Referring now to FIG. 3, there is shown a side view of feeder system **108** (partially schematically). Feeder system **108** is shown to include pickup assembly **111**, pickup tines **240**, pickup frame **241**, windguard mounting device **242**, and windguard assembly **128**. Windguard assembly **128** is shown in two orientations, with first arm **245** at an intermediate position (in solid lines, and substantially similar to the orientation of windguard assembly **128** in FIG. 2), and with first arm **245** rotated up (counter-clockwise on the page of FIG. 3) (in broken lines). Both orientations of windguard assembly **128** shown in FIG. 3 show windguard assembly **128** in normal operation and thus in the first position **237**, considering that second arm **246** has not been extended forward by actuator **251** (second arm **246** is fully retracted, in retraction position **257**) and thus also tine mechanism **248** remains in its down position **267**; being in the first position **237**, windguard assembly **128** (in either orientation in FIG. 5) is not yet positioned for a plug of crop material **136** to be cleared. With regard to the orientation of windguard assembly **128** shown in FIG. 3 with first arm **245** in the intermediate position (windguard assembly **128** in solid lines), second arm **246** is retracted and resting on stop mechanism **243**. Because second arm **246** is on stop mechanism **243**, chain **261** can be slack (alternatively, when second arm **246** is not resting on stop mechanism **243**, chain **261** can be taut, such that windguard assembly **128** can be, at least in part, suspended by chain **261**, if windguard assembly **128** is not otherwise supported by or resting on crop material **136** instead). First arm **245** includes proximal end **352** and distal end **355**. Actuator **251** includes a proximal end **366** which is pivotably coupled with proximal end **352** of first arm **245** (alternatively, actuator **251** could be reversed, with its proximal end **352** being pivotably coupled with second arm **246**, rather than proximal end **352** of first arm **245**). With second arm **246** being fully retracted, tine mechanism **248** is in its down position **267**, with tines **260** being down. FIG. 3 thus depicts a normal operating position of windguard assembly **128**, with windguard assembly **128** at resting height dictated by stop mechanism **243** (or, if not stop mechanism **243**, then the length of chain **261**). Further, each of connections **265A-H** is a pivotal connection, such that rotation with respect to each connection **265A-G** occurs about a respective axis of rotation that is perpendicular to the page of FIG. 3. This first orientation (in solid lines) of first arm **245** can occur when no or a relatively small quantity of crop material **136** is flowing through feeder system **108**. The pivot of first arm **245** at connection **265A** is generally free-floating, in the sense that first arm **245** can pivot at connection **265A** about a rotational axis extending between connections **265A** depending upon the quantity of crop material **136**. A larger quantity of crop material **136** pushes first arm **245** up so as to rotate a greater degree counter-clockwise at connection **265A**, and, conversely, a lesser quantity of crop material **136** enables first arm **245** to rotate a greater degree clockwise at connection **265A** (so long as this rotation clockwise is not blocked by stopping mechanism **243** of the length of chain **261**). In this way, windguard assembly **128** can be said to swing or be suspended by chain **261**. With regard to the orientation of windguard assembly **128** shown in FIG.

3 with first arm 245 rotated up (windguard assembly 128 in broken lines), this position of first arm 245 can occur when a relatively large quantity of crop material 136 is flowing through feeder system 108 (and thus tines 260 generally ride on a top portion of the mat of crop material 136), or a plug of crop material 136 has plugged or become jammed in feeder system 108. In this latter orientation (first arm 245 rotated up), first arm 245 can be at maximum rotation counter-clockwise, such that windguard assembly 128 is at maximum height (being supported by the crop material in this orientation, chain 261 can be slack). Thus, first arm 245 can freely pivot at connection 265A, depending upon the amount of crop material 136 in feeder system 108 (whether crop material 136 is not plugged, or is plugged, in feeder system 108).

(28) With further reference to FIG. 3, control system 122 is shown according to one embodiment of the present invention, that is, to include position sensors 335A, 335B relative to windguard assembly 128 (alternatively, the present invention is not required to be used in conjunction with sensors, such as position sensors 335A, 335B, but the present invention can function by user controlling, for example, hydraulic remotes/valves manually with sensors being absent). Thus, control system 122 can further include not only controller 123 but also sensor 335A and/or sensor 335B (additional sensors can be employed, such as additional position sensors for sensing a position of windguard assembly 128), each of which can be position sensors (for illustrative purposes controller 123 and sensors 335A, 335B are shown only in FIG. 3, but they can be impliedly present in other figures as well). Sensors 335A, 335B, each of which are schematically shown and positioned, are operatively coupled with controller 123 (and also controller 115) and are each configured to help determine whether windguard assembly 128 is in the first position 237 or the second position 438. Sensor 335A can be positioned to sense a position of the rod of actuator 251, a position of second arm 246 relative to first arm 245, and/or a position of first linkage 263 relative to first arm 245 (alternatively, sensor 335A, or another position sensor, could be placed inside the cylinder of actuator 251 to sense a position of the piston within the cylinder). The position of the rod of actuator 251 corresponds with the position of second arm 246, indicating whether second arm 246 is fully retracted, fully extended (and thus forward), or at an intermediate position therebetween. Further, in sensing the position of the rod, this sensing can be done by sensing a position of the pin or shaft of connection 265H within slot 254. In sensing the position of second arm 246 relative to first arm 245, sensor 335A can sense a linear and/or angular position of second arm 246 relative to first arm 245. In sensing the position of first linkage 263 relative to first arm 245, sensor 335A can sense, for example, an angular position of first linkage 263 relative to first arm 245. Sensor 335B can be positioned to sense a position of second linkage 264 relative to first linkage 263, a position of second linkage 264 and/or bar 259 relative to second arm 246, and/or a position of tines 260 relative to second arm 246. In sensing the position of second linkage 264 relative to first linkage 263, sensor 335B can be positioned on or near first or second linkage 263, 264 and can be configured to sense an angular position of second linkage 264 relative to first linkage 263. In sensing the position of second linkage 264 and/or bar 259 relative to second arm 246, sensor 335B can sense an angular position of second linkage 264 and/or bar 259 relative to second arm 246. In sensing the position of tines 260 relative to second arm 246, sensor 335B can sense a linear or angular position of tines 260 relative to second arm 246. Upon sensing any of these respective positions, sensor(s) 335A, 335B can output a respective position signal to controller 123. Controller 123 is configured for receiving the respective position signals from sensors 335A, 335B. Upon receiving these respective position signals, controller 123 can output position information to controller 115 based at least in part on these position signals, which can output this information to user by way of input/output device 120. User (i.e., operator of tractor 100) can enter a command in cab 105 via input/output device 120, and controller(s) 115 and/or 123 can cause windguard assembly 128 to adjust its position, in accordance with the user's command. Adjustment of the positioning of windguard assembly 128 can occur while baler 101 is or is not traversing across the ground, depending upon the circumstances, though removal of a plug of crop

material **136** can be typically performed when baler **101** is not traversing the ground. User's command can thus move windguard assembly into first position **237** and second position **438**, depending upon the circumstances. Thus, the present invention provides for in-cab adjustment of windguard assembly **128** to remove a plug of crop material **136**. Further, though normal use of windguard assembly **128** can be (as discussed herein) with second arm **246** fully retracted, a user may find it advantageous to run windguard assembly **128** with second arm **246** at least partially extended during normal operation of baler **101** (when baler **101** traverses the ground so as to pick up a windrow of crop material **136**), depending, for instance, on crop types and conditions; such a configuration of windguard assembly **128** can be controlled by user from cab **105** by way of input/output device **120**. Alternatively or in addition thereto, user may find it advantageous to adjust windguard assembly **128** with second arm **246** being at least partially extended, not necessarily while baler **101** is traversing the ground, but when baler **101** is stopped for other reasons than for clearing a crop plug, such as for maintenance of feeder system **108** or any other aspect of baler **101**. That is, extending second arm **246** and thus also moving tine mechanism **248** upwardly can create additional space within feeder system **108** and thereby improve accessibility to feeder system **108**, such as for maintenance of pickup assembly **111**, rotary feeder assembly **121**, windguard assembly **128**, and/or for replacement of tines **240** or hoops (i.e., which tines **240** pass between).

(29) Alternatively or in addition thereto, control system **129** can be configured to automatically unplug feeder system **108** of a plug of crop material **136**, without the user having to enter a command to do so (this alternative can occur with a user operating tractor **100** pulling baler **101**, or with autonomous tractor and/or baler operations). Though not shown, control system **122** of baler **101** can include at least one sensor configured for: (a) sensing a plug of crop material **136** in feeder system **108** (for example, optical sensors detecting a profile of crop material **136** indicative of a plug of crop material, and/or weight sensors detecting a weight of crop material **136** indicative of a plug of crop material **136**, and/or any other sensor(s) suitable for detecting a plug of crop material **136**); and (b) outputting a plug signal corresponding to the plug sensed by the sensor. Controller **123** (and/or controller **115**) is configured for: (a) receiving this plug signal; (b) for outputting an adjustment signal, based at least in part on the plug signal, so as to adjust windguard assembly **128** so as to remove the plug of crop material **136** and thereby to move windguard assembly to second position **438**; and (c) after removal of the plug, for outputting an additional adjustment signal, based at least in part upon another position signal indicative of windguard assembly **128** being in second position **438** and another signal indicative of the removal of the plug of crop material (i.e., another optical signal indicating a profile, or lack thereof, of crop material **136**, or a weight or mass of crop material **136**, indicating an absence of the plug), for moving windguard assembly **128** back to first position **237** for normal operation of baler **101**.

(30) Referring now to FIG. **4**, there is shown a side view of feeder system **108**, partially schematically. Feeder system **108** is shown to include pickup assembly **111** with tines **240**, and windguard assembly **128**. The primary difference between FIGS. **3** and **4** is that windguard assembly **128** is in the second position **438** in FIG. **4**, considering that second arm **246** has been extended forward by actuator **251** (second arm **246** is fully extended to extended position **458**) and thus also tine mechanism **248** is now in its up position **468**. Being in the second position **538**, windguard assembly **128** is positioned for a plug of crop material **136** to be cleared and thus is not in normal operating position for normal operation of baler **101** (that is, when crop material **136** is flowing into bale chamber **109**). Hydraulic cylinder assembly **251** has retracted its rod, pulling back the pin or shaft of connection **265H** to a proximal end of slot **254**, which has caused: (a) second arm **246** to pivot counter-clockwise (viewed from a bottom left of the page of FIG. **4**) at connection **265B** so that roller **247** extends farther forward; and (b) first linkage **263** has pivoted at connection **265E** and has pushed second linkage **264** forward at connection **265F**, which has caused second linkage **264** to pivot bar **259** at connection **265D**, resulting in tines **260** moving from their

down position **267** (which is their downmost position) to their up position **468** (which is their upmost position). The movement of second arm **246** and roller **247** forward creates space for a plug of crop material **136** to be backed out of and removed from feeder system **108**, and the movement of tines **260** to their up position **468** pivots the tip of tines **260** far enough up so that when crop material **136** is backed out (such as by reversing the normal direction **134** of rotor **144**) the tips of tines **260** do not stick into crop material **136** (as barbs) and thereby inhibit the removal of crop material **136** from feeder system **108**. Further, relative to FIG. **3**, first arm **245** has pivoted clockwise (and thus farther downward) at connection **265A**. Thus, being in second position **438**, the crop material **136** can be reversed (in direction **133**, FIG. **1**), which can occur, for example, when rotors **144** are reversed so as to remove a plug of crop material **136**. Thus, second position **438** is associated with a reversed operation of rotors **144**.

(31) Further, though not shown in FIG. **4**, first arm **245** can pivot at connection **265A**, as shown to occur in FIG. **3**. FIG. **4** shows first arm **245** in a downmost position, with first and/or second arm **245**, **246** resting on stop mechanism **243**; when arm(s) **245**, **246** do not rest on stop mechanism **243**, chain **261** can be taut, suspending at least portions of windguard assembly **128** (in this way, the height of windguard assembly **128** can depend upon the amount of crop material **136** (whether as a plug or not) in feeder system **108**, the height of stop mechanism **243**, or the length of chain **261**). This orientation of first arm **245** (shown in FIG. **4**) can occur when no or a relatively small quantity of crop material **136** is flowing through feeder system **108**. However, first arm **245** can freely pivot at connection **265A**, depending upon the amount of crop material **136** in feeder system **108**. Thus, first arm **245** can pivot at connection **265A** upward (counter-clockwise) due to an increased amount of crop material **136** in feeder system **108**. Thus, tines **260** can generally ride on a top portion of a mat of crop material **136**, or a plug of crop material **136** that has plugged or jammed in feeder system **108**. For high volume plugs of crop material **136**, windguard roller **247** may roll over the top of the plug during extension of second arm **246** by actuator **251**, in which case support chains **261** can be slack.

(32) In use, with respect to the embodiment of the present invention shown in FIGS. **2-6**, windguard assembly **128** can be run in first position **237** during normal operation of baler **101**. Upon a plug of crop material **136** forming in feeder system **108**, user (i.e., in cab **105** of tractor **100**) can enter a command via device **120** for windguard assembly **128** to move from first position **237** to second position **438** by way of control system **129**. Upon doing so, rotors **144** can be reversed (either by the same command or by way of an additional command) so that crop material **136** is reversed (in direction **133**) and thereby backed out of feeder system **108**. After removing the plug of crop material **136** (either by the same command or by way of an additional command), windguard assembly **128** can be returned to first position **237**, so that baling of crop material **136** can continue. Alternatively, this removal of crop material, and return to normal operation of baler **101**, can be performed automatically by control system **129**.

(33) Referring now to FIGS. **5-6**, there is shown a second embodiment of the present invention. All prior reference numbers remain the same or are increased by a multiple of 100 and are substantially similar to the structures and function of the embodiment shown in FIGS. **2-4**, unless otherwise shown and/or described; thus, in an effort to avoid repetition, what is shown or described with respect to FIGS. **2-4** applies to FIGS. **5-6**, unless otherwise shown or described differently with respect to FIGS. **5-6** (FIG. **1** is generic to both embodiments shown in FIGS. **2-6**). The primary difference between the first embodiment and the second embodiment is the windguard assembly.

(34) Referring now to FIG. **5**, there is shown a perspective view of baler **501**, with portions broken away. Baler **501**, which can be pulled by tractor **100**, includes baler control system **522**, which includes controller **123** and sensor(s) **635A**, **635B**, and baler control system **522** is a part of unified control system **529**, which also includes tractor control system **114**. Further, baler **501** includes feeder system **508**. Feeder system **508** includes pickup assembly **111**, rotary feeder assembly **121**, and windguard assembly **528**, which is shown in first position **537**. Pickup assembly **111** includes

pickup tines **240**, pickup frame **241**, windguard mounting device **242**, and stop mechanism **243**, substantially as described above. Rotary feeder assembly **121** includes rotors **144**, substantially as described above. Windguard assembly **528** includes first arm **545**, second arm **546**, roller **547**, tine mechanism **548**, chain assembly **549**, linkage assembly **550**, actuator **551**, and connections **565A-H** (each of which is a pivot connection). First arm **545** is pivotably coupled with windguard mounting device **242** at connection **565A** and includes proximal end **552**, distal end **555**, central section **569**, and first arm channel **553**. Second arm **546** is pivotably coupled with first arm **545** at connection **565B**, includes second arm channel **556**, and is shown in FIG. 5 in retracted position **557**, resting on stop mechanism **243** (if second arm **546** were not resting on stop mechanism **243**, then second arm **546** can be supported and thus suspended by chain **561**, or supported by crop material **136**). Roller **547** is pivotably connected with second arm **546** at connection **565C**. Tine mechanism **548** includes tine bar **559** and tines **560**, and bar **559** is pivotably coupled with second arm **546** at connection **565D**. With second arm **546** in retraction position **557** in FIG. 5, tine mechanism **548** is in down position **567**. Chain assembly **549** includes a chain **561** and can include a chain sleeve (not shown in FIG. 5). Linkage assembly **550** includes first linkage **563** (pivotably connected to first arm **545** at connection **565E**) and second linkage **564** pivotably coupled with first linkage at connection **565F**. Second linkage **564** is rigidly connected with tine bar **559**, such as by welding. Actuator **551** can be a hydraulic cylinder assembly with the cylinder, the piston, and the rod, though other types of actuators can be used (as described above). Actuator **551** includes a proximal end **566**, which can include a proximal end of the cylinder (as shown in FIG. 5). Proximal end **566** is pivotably connected to first arm **545** at connection **565G**; more specifically, proximal end **566** is pivotably coupled with central section **569** of first arm **545** at connection **565G**. Further, a distal end of actuator **551** can be formed by the rod and is pivotably coupled with second arm **546** at connection **565H**. Unlike the first embodiment, this connection **565H** does not involve sliding a pin or shaft of connection **565H** within a slot formed by first arm **545**. When the rod of actuator **551** extends, second arm **546** extends forward, causing second arm **546** to rotate counter-clockwise (viewed from the bottom left in the page of FIG. 5) about connection **565B**; conversely, when the rod of actuator **551** retracts, second arm **546** retracts backward, causing second arm **546** to rotate clockwise about connection **565B**. Further, as with the first embodiment, windguard assembly **528** can pivot about connection **565A** depending upon the quantity of crop material **136**.

(35) Referring now to FIG. 6, there is shown baler **501** with feeder system **508**, with windguard assembly **528** being in second position **638**. As such, second arm **546** is in extended position **658**, and tine mechanism **548** is in up position **668**. In second position **638**, windguard assembly **528** is configured for clearing of a crop plug. Rather than resting on stop mechanism **243**, second arm **546** can be suspended by a taut chain **561** (or, alternatively, be supported by crop material **136**, either as a plug or not). In FIG. 6, first arm **545** has rotated clockwise about connection **565A** (relative to FIG. 5). Further, as with the first embodiment, the extension of second arm **546** has caused first linkage **563** to push on second linkage **564**, which has caused tine bar **559**, and thus tines **560**, to rotate upward (clockwise). As in the first embodiment of the present invention, first arm **545** can rotate at pivot connection **565A** depending upon the quantity of crop material **136**. FIG. 6 shows windguard assembly **528** with first arm **545** pivoted up. This can occur when windguard assembly encounters a relatively large volume of crop material **136**, which can cause first arm **545** to pivot about an axis of rotation (perpendicular to the page of FIG. 6) associated with connection **565A**. Thus, though not shown in FIG. 6, first arm **545** can be pivoted downwards (clockwise) when encountering less crop material, and can be stopped by stop mechanism **243** and/or chain **561**. In use, feeder system **508** can be used substantially similarly to what is described with respect to feeder system **108**. Further, sensors **635A**, **635B** are substantially similar to sensors **335A**, **335B** and thus are also position sensors that send their data to controller **123** (and/or controller **115**), except that sensor **635A** does not sense positional information relative to a slot similar to slot **254**. Further, position sensors **635A**, **635B** are optional, such that the present invention can be used

without them. That is, as with the first embodiment a user can still enter a command (such as by way of device **120**) to clear a crop plug, even without information from sensors **635A**, **635B**.

(36) Further, yet another embodiment of the present invention is provided. That is, the windguard assembly of the present invention can be mounted to baler tongue **107**. Thus, any of the embodiments of the windguard assembly according to the present invention can be adapted so as to mount to tongue **107**. Further, in accordance with yet another embodiment of the present invention, the windguard assembly may omit the roller, or may have a baffle in place of tines.

(37) Referring now to FIG. 7, there is shown a flow diagram showing a method **790** of using an agricultural baler **101**, **501**, the method **790** including the steps of: providing **791** a baler frame **130** and a feeder system **108**, **508**, the feeder system **108**, **508**, being coupled with the baler frame **130** and including a windguard assembly **128**, **528**; occupying **792** selectively, by the windguard assembly **128**, **528**, a first position **237**, **537**, the first position **237**, **537** being associated with a normal flow direction **132** of a crop material **136** relative to the feeder system **108**, **508**; and occupying **793** selectively, by the windguard assembly **128**, **528** a second position **438**, **638**, the second position **438**, **638** being associated with a reversal **133** of the normal flow direction **132** of the crop material **136**. The feeder system **108**, **528** can include a pickup assembly **111** and a rotary feeder assembly **121** each of which is coupled with the baler frame **130**, the windguard assembly **128**, **528** being coupled with the pickup assembly **111**, the rotary feeder assembly **121** including a rotor device **144**, the second position **438**, **638** being associated with a reversed operation of the rotor device **144**. The windguard assembly **128**, **528** can include: a first arm **245**, **545** pivotably coupled with the pickup assembly **111**; a second arm **246**, **546** pivotably coupled with the first arm **245**, **545**; an actuator **251**, **551** pivotably coupled with the first arm **245**, **545** and the second arm **246**, **546** and configured for moving the second arm **246**, **546** between a retracted position **257**, **557** and an extended position **458**, **658**, the retracted position **257**, **557** being associated with the first position **237**, **537**, the extended position **458**, **658** being associated with the second position **438**, **638**; a tine mechanism **248**, **548** pivotably coupled with the second arm **246**, **546**; a linkage assembly **250**, **550** coupled with the first arm **245**, **545** and the tine mechanism **248**, **548** and configured for moving the tine mechanism **248**, **548** between a down position **267**, **567** and an up position **468**, **668** when the second arm **245**, **545** moves between the retracted position **257**, **557** and the extended position **458**, **658**, respectively. The first arm **245** can include a proximal end **352**, the actuator **251** including a proximal end **366** which is pivotably coupled with the proximal end **352** of the first arm **245**. The first arm **545** can include a central section **569**, the actuator **551** including a proximal end **552** which is pivotably coupled with the central section **569** of the first arm **545**.

(38) It is to be understood that the steps of method **790** may be performed by controller **115**, **123** upon loading and executing software code or instructions which are tangibly stored on a tangible computer readable medium, such as on a magnetic medium, e.g., a computer hard drive, an optical medium, e.g., an optical disc, solid-state memory, e.g., flash memory, or other storage media known in the art. Thus, any of the functionality performed by controller **115**, **123** described herein, such as the method **790**, is implemented in software code or instructions which are tangibly stored on a tangible computer readable medium. The controller **115**, **123** loads the software code or instructions via a direct interface with the computer readable medium or via a wired and/or wireless network. Upon loading and executing such software code or instructions by controller **115**, **123**, controller **115**, **123** may perform any of the functionality of controller **115**, **123** described herein, including any steps of the method **790**.

(39) The term “software code” or “code” used herein refers to any instructions or set of instructions that influence the operation of a computer or controller. They may exist in a computer-executable form, such as machine code, which is the set of instructions and data directly executed by a computer's central processing unit or by a controller, a human-understandable form, such as source code, which may be compiled in order to be executed by a computer's central processing unit or by

a controller, or an intermediate form, such as object code, which is produced by a compiler. As used herein, the term “software code” or “code” also includes any human-understandable computer instructions or set of instructions, e.g., a script, that may be executed on the fly with the aid of an interpreter executed by a computer's central processing unit or by a controller.

(40) These and other advantages of the present invention will be apparent to those skilled in the art from the foregoing specification. Accordingly, it is to be recognized by those skilled in the art that changes or modifications may be made to the above-described embodiments without departing from the broad inventive concepts of the invention. It is to be understood that this invention is not limited to the particular embodiments described herein, but is intended to include all changes and modifications that are within the scope and spirit of the invention.

Claims

1. A windguard assembly for a feeder system of an agricultural baler, the agricultural baler including a baler frame, the feeder system being coupled with the baler frame, the feeder system including a pickup assembly, the windguard assembly comprising: a first arm configured to be pivotably coupled to the pickup assembly; a second arm pivotably coupled to the first arm; a roller rotatably coupled to the second arm; a tine mechanism pivotably coupled with the second arm; an actuator pivotably coupled with the first arm and the second arm and configured for moving the second arm between a retracted position associated with a first position of the roller and an extended position associated with a second position of the roller, the first position being associated with a normal flow direction of a crop material relative to the feeder system, the second position being associated with a reversal of the normal flow direction of the crop material; and a linkage assembly coupled with the first arm and the tine mechanism to move the tine mechanism between a down position and an up position when the second arm moves between the retracted position and the extended position, respectively.
2. The windguard assembly of claim 1, wherein the second position is associated with a reversed operation of a rotor device of a rotary feeder assembly, which is coupled with the baler frame.
3. The windguard assembly of claim 1, wherein the first arm includes a proximal end, the actuator including a proximal end which is pivotably coupled with the proximal end of the first arm.
4. The windguard assembly of claim 1, wherein the first arm includes a central section, the actuator including a proximal end which is pivotably coupled with the central section of the first arm.
5. An agricultural machine, comprising: a machine frame; and a feeder system coupled with the machine frame and including a pickup assembly and a windguard assembly, the windguard assembly comprising: a first arm configured to be pivotably coupled to the pickup assembly; a second arm pivotably coupled to the first arm; a roller rotatably coupled to the second arm; a tine mechanism pivotably coupled with the second arm; an actuator pivotably coupled with the first arm and the second arm and configured for moving the second arm between a retracted position associated with a first position of the roller and an extended position associated with a second position of the roller, the first position being associated with a normal flow direction of a crop material relative to the feeder system, the second position being associated with a reversal of the normal flow direction of the crop material; and a linkage assembly coupled with the first arm and the tine mechanism to move the tine mechanism between a down position and an up position when the second arm moves between the retracted position and the extended position, respectively.
6. The agricultural machine of claim 5, wherein the agricultural machine is an agricultural baler and the machine frame is a baler frame, the feeder system further including a rotary feeder assembly coupled with the baler frame, the rotary feeder assembly including a rotor device, the second position being associated with a reversed operation of the rotor device.
7. The agricultural machine of claim 5, wherein the first arm includes a proximal end, the actuator including a proximal end which is pivotably coupled with the proximal end of the first arm.

8. The agricultural machine of claim 5, wherein the first arm includes a central section, the actuator including a proximal end which is pivotably coupled with the central section of the first arm.
9. A method of using an agricultural baler, the method comprising the steps of: providing a baler frame and a feeder system, the feeder system being coupled with the baler frame and including a pickup assembly and a windguard assembly, the windguard assembly comprising a first arm configured to be pivotably coupled to the pickup assembly, a second arm pivotably coupled to the first arm, a roller rotatably coupled to the second assembly, a tine mechanism pivotably coupled with the second arm, an actuator pivotably coupled with the first arm and the second arm and configured for moving the second arm between a retracted position associated with a first position of the roller and an extended position associated with a second position of the roller, the first position being associated with a normal flow direction of a crop material relative to the feeder system, the second position being associated with a reversal of the normal flow direction of material, and a linkage assembly coupled with the first arm and the tine mechanism to move the tine mechanism between a down position and an up position when the second arm moves between the retracted position and the extended position, respectively; occupying selectively, by the roller, a first position, with the tine mechanism in the down position, the first position being associated with a normal flow direction of a crop material relative to the feeder system by moving the second arm into the retracted position; and occupying selectively, by the roller, a second position, with the tine mechanism in the up position, the second position being associated with a reversal of the normal flow direction of the crop material by moving the second arm into the extended position.
10. The method of claim 9, wherein the feeder system includes a pickup assembly and a rotary feeder assembly each of which is coupled with the baler frame, the windguard assembly being coupled with the pickup assembly, the rotary feeder assembly including a rotor device, the second position being associated with a reversed operation of the rotor device.
11. The method of claim 9, wherein the first arm includes a proximal end, the actuator including a proximal end which is pivotably coupled with the proximal end of the first arm.
12. The method of claim 9, wherein the first arm includes a central section, the actuator including a proximal end which is pivotably coupled with the central section of the first arm.
-